

SGA 3, Exposé VIII
Diagonalizable Groups
English Reconstruction

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Exposé VIII

Diagonalizable Groups

by A. Grothendieck⁰

1. Biduality

Let $\mathcal{C}$ be a category, which as usual we identify with a full subcategory of

$$\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}^\circ, \mathbf{Set})$$

(cf. Exposé I). Let I be a commutative group functor, that is, an object of $\widehat{\mathcal{C}}$ endowed with a commutative group structure (cf. I, 2.1).¹ For every $X \in \text{Ob}(\widehat{\mathcal{C}})$, the object $\text{Hom}(X, I)$ carries the commutative group structure induced by that of I . For every group G in $\widehat{\mathcal{C}}$, put

$$D(G) = \text{Hom}_{\text{gr}}(G, I),$$

the subobject of $\text{Hom}(G, I)$ defined, for every $S \in \text{Ob}(\mathcal{C})$, by

$$(x) \quad D(G)(S) = \text{Hom}_{S\text{-gr}}(G_S, I_S),$$

where $G_S = G \times S$ and $I_S = I \times S$ are regarded as S -groups, that is, as groups in $\widehat{\mathcal{C}}/S$. Thus $D(G)$ is a $\widehat{\mathcal{C}}$ -subgroup of $\text{Hom}(G, I)$. In this way we obtain a contravariant functor D from the category of $\widehat{\mathcal{C}}$ -groups to the category of commutative $\widehat{\mathcal{C}}$ -groups.

The right-hand side of (x) may also be interpreted as the subset of $\text{Hom}(G \times S, I)$ consisting of morphisms $G \times S \rightarrow I$ that are “multiplicative in the first argument G .” Moreover, the preceding formulas remain valid when S is an arbitrary object of $\mathcal{C}$, not necessarily one arising from $\mathcal{C}$.

Now take for S a group in $\widehat{\mathcal{C}}$, denoted G' . In the left-hand side $\text{Hom}(G', D(G))$ of (x), consider the subset $\text{Hom}_{\text{gr}}(G', D(G))$ consisting of morphisms that respect the group structures of G' and $D(G)$.

It corresponds to the subset of $\text{Hom}(G \times G', I)$ consisting of morphisms that are multiplicative in each of the two arguments—which one may call *bilinear morphisms from $G \times G'$ to I* , or *pairings of G and G' with values in I* . We therefore have

$$(xx) \quad \text{Hom}_{\text{gr}}(G', D(G)) \xrightarrow{\sim} \text{Hom}_{\text{bil}}(G \times G', I),$$

⁰Editor’s note: version 1.1 of November 8, 2009. Additions were made in 1.2, 1.4, 1.7, 3.1, 3.4, 4.5.1, 6.4, and 6.8; 1.5.1 and Section 7 remain to be revised.

¹Editor’s note: the original has been expanded in what follows.

functorially in the pair (G, G') . Since the right-hand side is symmetric in G and G' , this gives a functorial bijection

$$(xxx) \quad \mathrm{Hom}_{\mathrm{gr}}(G', D(G)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(G, D(G')).$$

In other words, giving a group homomorphism $G' \rightarrow D(G)$ “amounts to the same thing” as giving a group homomorphism $G \rightarrow D(G')$: both amount to giving a pairing $G \times G' \rightarrow I$.

Apply this with $G' = D(G)$ and with the identity homomorphism $G' \rightarrow D(G)$. We obtain a canonical homomorphism

$$(xxxx) \quad G \longrightarrow D(D(G)).$$

Definition 1.0.² *The group G is called reflexive (relative to I) if the preceding homomorphism is an isomorphism. This implies, in particular, that G is commutative.*

We therefore have:

Proposition 1.0.1. *The functor D induces an anti-equivalence from the category of reflexive $\mathcal{C}$ -groups to itself.*

In particular, if G and H are reflexive groups, D induces an isomorphism

$$\mathrm{Hom}_{\mathrm{gr}}(G, H) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(D(H), D(G)).$$

It is enough, in fact, that H be reflexive, as follows from (xxx).

Definition 1.0.2. *As usual, a $\mathcal{C}$ -group G is called reflexive if it is reflexive as a $\widehat{\mathcal{C}}$ -group, regardless of whether $D(G)$ is representable.*

Thus D gives an anti-equivalence from the category of reflexive $\mathcal{C}$ -groups G for which $D(G)$ is representable to itself.

Remark 1.0.3. To conclude these generalities, note that formation of the dual $D(G)$ commutes with base extension, which therefore carries reflexive groups to reflexive groups.

We shall henceforth work in the following setting. Let $\mathcal{C} = (\mathbf{Sch})_S$ be the category of preschemes over S , and let $I = \mathbf{G}_{m,S}$ be the “multiplicative group over S ” (cf. Exposé I). For every ordinary group M , consider the constant S -group M_S . For every group prescheme J over S , there is a canonical isomorphism, functorial in M and J and compatible with base extension,

$$\mathrm{Hom}_{S\text{-gr}}(M_S, J) = \mathrm{Hom}_{\mathrm{gr}}(M, J(S)).$$

Applying this with $J = I = \mathbf{G}_{m,S}$ and with a variable S' over S , we obtain a functorial isomorphism

$$(1.0.4) \quad D(M_S)(S') \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(M, \mathbf{G}_m(S')).$$

We thus recover the functor already considered in I, 4.4, also denoted $D_S(M)$, which is representable when M is commutative, since

$$D_S(M) = D(M_S) = \mathrm{Spec} \mathcal{O}_S(M),$$

where $\mathcal{O}_S(M)$ denotes the group algebra of M over $\mathcal{O}_S$. In general, moreover, $D(M)$ is unchanged when M is replaced by its abelianization, so no information is lost by assuming that M is commutative.

²Editor’s note: the numbering 1.0, 1.0.1, and so on has been added to make the definitions and statements contained here explicit.

Definition 1.1. A group prescheme G over S is called diagonalizable if it is isomorphic to a scheme of the form

$$D_S(M) = D(M_S) = \operatorname{Hom}_{S\text{-gr}}(M_S, \mathbf{G}_m)$$

for a suitable commutative group M . It is called locally diagonalizable if every point of S has an open neighborhood U such that $G|_U$ is diagonalizable.

Theorem 1.2. Let Γ be a constant commutative group scheme over S ; that is, suppose that it is isomorphic to a group scheme of the form M_S , where M is an ordinary commutative group. Then Γ is reflexive: the canonical homomorphism

$$\Gamma \longrightarrow D(D(\Gamma))$$

is an isomorphism.³ Hence the diagonalizable group $D(M_S)$ is also reflexive.

In view of the definitions, this follows from the next statement, applied to a prescheme S' over S .

Corollary 1.3. Let $G = D(M_S)$. Every homomorphism of S -groups

$$u : G \longrightarrow \mathbf{G}_{m,S}$$

is defined by a unique section of M_S over S , equivalently by a unique locally constant map from S to M .

Proof. By definition,

$$\mathbf{G}_{m,S} = \operatorname{GL}(1)_S = \operatorname{Aut}_{\mathcal{O}_S\text{-mod}}(\mathcal{O}_S).$$

Thus giving a group homomorphism $G \rightarrow \mathbf{G}_{m,S}$ is equivalent to giving $\mathcal{O}_S$ the structure of a G - $\mathcal{O}_S$ -module compatible with its natural $\mathcal{O}_S$ -module structure (cf. I, 4.7). By I, 4.7.3, this is also equivalent to giving an M -grading on $\mathcal{O}_S$, that is, a direct sum decomposition

$$\mathcal{O}_S = \bigoplus_{m \in M} \mathcal{L}_m.$$

It is well known that a direct summand of a locally free module of finite type is locally free of finite type. Hence, near each point of S , every $\mathcal{L}_m$ is either zero or free of rank one; in the latter case it identifies with $\mathcal{O}_S$ on that neighborhood. Let S_m be the open subset of S on which the latter alternative occurs. The direct-sum decomposition shows that the S_m cover S and are pairwise disjoint.

Consequently, giving a group homomorphism $G \rightarrow \mathbf{G}_{m,S}$ is equivalent to giving a decomposition of S as a union of pairwise disjoint open subsets S_m , $m \in M$, hence to giving a locally constant map from S to M . This proves Corollary 1.3 and therefore Theorem 1.2. $\square$

Corollary 1.4. A diagonalizable group is reflexive, and therefore so is a locally diagonalizable group. If M and N are ordinary commutative groups, the natural homomorphism

$$\operatorname{Hom}_{S\text{-gr}}(M_S, N_S) \longrightarrow \operatorname{Hom}_{S\text{-gr}}(D(N_S), D(M_S))$$

is bijective.

³Editor's note: the sentence that follows has been added.

⁴ Since the preceding isomorphism is compatible with base extension, it yields an isomorphism of S -functors in groups

$$(1) \quad \underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S)).$$

For every S -prescheme T ,

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S)(T) = \mathrm{Hom}_{\mathrm{gr}}(M, \Gamma(N_T/T)),$$

and by I, 1.8, $\Gamma(N_T/T)$ is the abelian group of locally constant maps $T \rightarrow N$. On the other hand, let $\mathrm{Hom}_{\mathrm{gr}}(M, N)_S$ be the constant S -group associated with the ordinary abelian group $\mathrm{Hom}_{\mathrm{gr}}(M, N)$. There is an evident homomorphism of S -functors in commutative groups

$$(2) \quad \mathrm{Hom}_{\mathrm{gr}}(M, N)_S \xrightarrow{\theta} \underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S),$$

which is always a monomorphism and is an isomorphism if M is of finite type.⁵

Part (a) of the following corollary follows from the preceding discussion; part (b) follows from it by the descent results recalled in 1.7.⁶

Corollary 1.5.

- (a) *Let M and N be ordinary commutative groups, with M of finite type. Then there is an isomorphism*

$$\mathrm{Hom}_{\mathrm{gr}}(M, N)_S \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S));$$

consequently, $\underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S))$ is representable.

- (b) *More generally, if G and H are locally diagonalizable and H is of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr}}(G, H)$ is representable.*

⁷ Corollary 1.5 gives:

Corollary 1.6. *Under the hypotheses of 1.5, if S is connected, then*

$$\mathrm{Hom}_{S\text{-gr}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(M, N)$$

and

$$\mathrm{Isom}_{S\text{-gr}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Isom}_{\mathrm{gr}}(M, N).$$

⁴Editor's note: the discussion that follows has been expanded.

⁵Editor's note: write $F(M)$ and $G(M)$ for the left- and right-hand sides of (2), respectively. If $M = M_1 \oplus M_2$, there are canonical isomorphisms $F(M) = F(M_1) \oplus F(M_2)$ and $G(M) = G(M_1) \oplus G(M_2)$. It is therefore enough to check that θ is an isomorphism for $M = \mathbf{Z}/r\mathbf{Z}$, $r \geq 0$. In that case $F(M) = ({}_rN)_S$, where ${}_rN$ is the kernel of $r \mathrm{id}_N$, and for every $T \rightarrow S$ the homomorphism

$$F(M)(T) = \Gamma({}_rN_T/T) \longrightarrow G(M)(T) = {}_r\Gamma(N_T/T)$$

is bijective.

⁶Editor's note: the original stated after 1.5 that, more generally, if G and H are locally diagonalizable and H is of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr}}(G, H)$ is representable. That assertion has been included in 1.5 and its proof made explicit in 1.7.

⁷Editor's note: a statement 1.5.1 treating the functor $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, H)$, considered in X, 5.10 and 5.11, should be added.

1.7. Descent of representability. ⁸ We recall here several descent results that will be used frequently below.

Scholium 1.7.1. *Let S be a prescheme and let X, Y, T be S -preschemes. If (T_i) is an open covering of T , and if $T_{ij} = T_i \cap T_j = T_i \times_T T_j$, then, because giving a morphism of T -preschemes $X_T \rightarrow Y_T$ is local on T , there is an exact sequence of sets*

$$(1) \quad \mathrm{Hom}_T(X_T, Y_T) \longrightarrow \prod_i \mathrm{Hom}_{T_i}(X_{T_i}, Y_{T_i}) \rightrightarrows \prod_{i,j} \mathrm{Hom}_{T_{ij}}(X_{T_{ij}}, Y_{T_{ij}}).$$

Thus $\underline{\mathrm{Hom}}_S(X, Y)$ is a local S -functor, that is, a sheaf on $(\mathbf{Sch})/S$ for the Zariski topology.

More generally, by IV, 4.5.13, $\underline{\mathrm{Hom}}_S(X, Y)$ is a sheaf on $(\mathbf{Sch})/S$ for every topology coarser than the canonical topology, for example the fpqc topology. If G and H are group preschemes over S , it follows that the subfunctor $\underline{\mathrm{Hom}}_{S\text{-gr}}(G, H)$ is an fpqc sheaf, hence a fortiori a local functor.

Lemma 1.7.2. ⁹ *Let F be a local S -functor.*

- (i) *Suppose that there is an open covering (S_i) of S such that the restriction $F_i = F \times_S S_i$ of F to each S_i is representable by an S_i -prescheme X_i . Then F is representable by an S -prescheme X .*
- (ii) *Suppose that F is an fpqc sheaf and that there is a faithfully flat quasi-compact morphism $S' \rightarrow S$ such that $F' = F \times_S S'$ is representable by an S' -prescheme X' . Then X' carries a descent datum relative to $S' \rightarrow S$ (cf. IV, 2.1). If this descent datum is effective—as it is when X' is affine over S' —then F is representable by an S -prescheme X .*

Proof. (i) The hypothesis implies that $X_i \times_S S_j$ and $X_j \times_S S_i$ both represent the restriction of F to $S_{ij} = S_i \times_S S_j$. The Yoneda lemma therefore gives a unique isomorphism of S_{ij} -preschemes

$$c_{ji} : X_i \times_S S_j \xrightarrow{\sim} X_j \times_S S_i.$$

Over $S_{ijk} = S_i \times_S S_j \times_S S_k$, we obtain the commutative diagram of isomorphisms

$$\begin{array}{ccc} X_i \times_S S_j \times_S S_k & \xrightarrow{c_{ji} \times \mathrm{id}_{S_k}} & X_j \times_S S_i \times_S S_k = X_j \times_S S_k \times_S S_i \\ \parallel & & \downarrow c_{kj} \times \mathrm{id}_{S_i} \\ X_i \times_S S_k \times_S S_j & \xrightarrow{c_{ki} \times \mathrm{id}_{S_j}} & X_k \times_S S_i \times_S S_j = X_k \times_S S_j \times_S S_i. \end{array}$$

Indeed, all of its objects represent the restriction of F to S_{ijk} . Thus the c_{ji} satisfy the usual cocycle relation $c_{kj} \circ c_{ji} = c_{ki}$.

It follows that the X_i glue to an S -prescheme X with $X \times_S S_i = X_i$ for every i . For every Y over S_i ,

$$(*) \quad F(Y) = F_i(Y) = \mathrm{Hom}_{S_i}(Y, X \times_S S_i) = \mathrm{Hom}_S(Y, X) = h_X(Y).$$

For an arbitrary $Y \rightarrow S$, the $Y_i = Y \times_S S_i$ form an open covering of Y ; put $Y_{ij} = Y_i \times_Y Y_j = Y \times_S S_{ij}$. Since F is local by hypothesis and h_X is local because the Zariski topology is

⁸Editor's note: this paragraph has been added to make explicit the “local on S ” character of the representability of certain sheaves on S , which is used repeatedly below and was used implicitly in the original.

⁹Editor's note: see also Remark XI, 3.4.

coarser than the canonical topology, both $F(Y)$ and $h_X(Y)$, in view of $(*)$, identify with the kernels of the respective double arrows

$$\prod_i F(Y_i) \rightrightarrows \prod_{i,j} F(Y_{ij}), \quad \prod_i h_X(Y_i) \rightrightarrows \prod_{i,j} h_X(Y_{ij}).$$

This proves (i).

(ii) Put $S'' = S' \times_S S'$, and let S''_1 and S''_2 denote S'' regarded as an S' -prescheme through the first and second projections. The functors $F''_1 = F' \times_{S'} S''_1$ and $F''_2 = F' \times_{S'} S''_2$ are represented respectively by $X''_1 = X' \times_{S'} S''_1$ and $X''_2 = X' \times_{S'} S''_2$. Since

$$F''_1 = F \times_S S'' = F''_2,$$

there is a unique S'' -isomorphism

$$c : X''_1 \xrightarrow{\sim} X''_2.$$

Let q_i , respectively p_{ji} , be the projection from $S''' = S' \times_S S' \times_S S'$ onto its i -th factor, respectively onto its i -th and j -th factors. Let $X'''_i = X' \times_{S'} S'''_i$, where S'''_i means S''' viewed as an S' -prescheme through q_i , and let $p_{ji}^*(c) : X'''_i \xrightarrow{\sim} X'''_j$ be obtained from c by base change. We obtain the diagram

$$\begin{array}{ccc} X'''_1 & \xrightarrow{p_{21}^*(c)} & X'''_2 \\ & \searrow p_{31}^*(c) & \downarrow p_{32}^*(c) \\ & & X'''_3. \end{array}$$

All three objects represent the restriction of F to S''' , so the diagram commutes:

$$p_{32}^*(c) \circ p_{21}^*(c) = p_{31}^*(c).$$

Thus c is a descent datum on X' relative to $S' \rightarrow S$ (cf. IV, 2.1).

Suppose that this descent datum is effective, so there is an S -prescheme X such that $X' \simeq X \times_S S'$. This is the case when X' is affine over S' , by SGA 1, VIII, 2.1.¹⁰ For every $Y \rightarrow S'$, one then has

$$(**) \quad F(Y) = F'(Y) = \operatorname{Hom}_{S'}(Y, X \times_S S') = \operatorname{Hom}_S(Y, X) = h_X(Y).$$

For arbitrary $Y \rightarrow S$, put $Y' = Y \times_S S'$ and $Y'' = Y' \times_Y Y' \simeq Y \times_S S''$. Like $S' \rightarrow S$, the morphism $Y' \rightarrow Y$ is faithfully flat and quasi-compact, hence an $\mathcal{M}$ -effective epimorphism, where $\mathcal{M}$ is the family of faithfully flat quasi-compact morphisms. Thus the equivalence relation

$$Y' \times_Y Y' \rightrightarrows Y'$$

is $\mathcal{M}$ -effective and has quotient Y . Since F is an fpqc sheaf by hypothesis, while h_X is one because the fpqc topology is coarser than the canonical topology, both $F(Y)$ and $h_X(Y)$, in view of $(**)$, identify with the respective kernels of

$$F(Y') \rightrightarrows F(Y' \times_Y Y'), \quad h_X(Y') \rightrightarrows h_X(Y' \times_Y Y').$$

This proves (ii). □

¹⁰Editor's note: for another effectivity criterion, see X, 5.4–5.6.

Corollary 1.7.3. *Let F be an fpqc sheaf on $(\mathbf{Sch})/S$. Suppose there is an open covering (S_i) of S , and for each i a faithfully flat quasi-compact morphism $S'_i \rightarrow S_i$, such that $F'_i = F \times_S S'_i$ is representable by an S'_i -prescheme X'_i affine over S'_i . Then F is representable by an S -prescheme X affine over S , with $X \times_S S'_i = X'_i$ for every i . If, moreover, every $X'_i \rightarrow S'_i$ is a closed immersion (respectively, a finite étale morphism), then so is $X \rightarrow S$.*

The first assertion follows from 1.7.2. For the second, it is enough to check that every morphism $X \times_S S_i \rightarrow S_i$ is a closed immersion (respectively, finite and étale); this follows from EGA IV₂, 2.7.1 (respectively, together with IV₄, 17.7.3).

Remark 1.7.4. Assertion 1.5(b) follows, as announced, from 1.7.1 and 1.7.2(i).

2. Scheme-theoretic Properties of Diagonalizable Groups

These properties are summarized in the following proposition.

Proposition 2.1. *Let S be a nonempty prescheme, let M be an ordinary commutative group, and let $G = D(M_S)$ be the diagonalizable S -group defined by M . Then:*

- (a) *G is faithfully flat over S and affine over S (a fortiori, quasi-compact over S);*
- (b) *M is of finite type if and only if G is of finite type over S , if and only if G is of finite presentation over S ;*
- (c) *M is finite if and only if G is finite over S , if and only if G is of finite type over S and is annihilated by an integer $n > 0$. In that case,*

$$\deg(G/S) = \text{Card}(M);$$

- (c') *M is a torsion group if and only if G is integral over S ;*
- (d) *$M = 0$ if and only if G is the unit S -group;*
- (e) *M is of finite type and the order of its torsion subgroup is prime to the residue characteristics of S if and only if G is smooth over S .*

The verification of (a) through (d) is immediate and is left to the reader. Let us prove (e). If G is smooth over S , it is locally of finite presentation over S , hence of finite presentation over S , since it is affine over S ; thus M is of finite type. We may therefore assume from the outset that M is of finite type, and hence that G is of finite presentation over S . Then¹¹ G is smooth over S if and only if its geometric fibers are smooth, reducing us to the case in which S is the spectrum of an algebraically closed field k .

Write $M = T \times L$, where T is the torsion subgroup and L is free, with $L \simeq \mathbb{Z}^r$. Then

$$D(M) = D(T) \times D(L), \quad D(L) \simeq \mathbb{G}_m^r,$$

and $D(L)$ is smooth over k . Thus $G = D(M)$ is smooth over k if and only if $D(T)$ is smooth. Since $D(T)$ is finite over k of degree equal to the order n of T , this means that $D(T)(k)$ has n elements. Now T is isomorphic to a sum of groups $\mathbb{Z}/n_i\mathbb{Z}$, where $n = \prod_i n_i$; consequently

$$D(T) = \prod_i D(\mathbb{Z}/n_i\mathbb{Z}) = \prod_i \mu_{n_i},$$

¹¹Editor's note: this uses the fact that G is flat over S , by (a).

where μ_{n_i} is the group scheme of n_i -th roots of unity. Hence

$$\text{card}(D(T)(k)) = \prod_i \text{card}(\mu_{n_i}(k)).$$

Here $\text{card}(\mu_{n_i}(k))$, the number of n_i -th roots of unity in k , is at most n_i , and equality holds if and only if n_i is prime to the characteristic p of k . [The source prints $\mu_i(k)$ in both occurrences.] It follows that

$$\text{card}(D(T)(k)) = n, \quad n = \prod_i n_i,$$

if and only if every n_i is prime to p , that is, if and only if n is prime to p . $\square$

3. Exactness Properties of the Functor D_S

Theorem 3.1. *Let S be a prescheme, and let*

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be an exact sequence of ordinary commutative groups. Consider the sequence of transposed homomorphisms

$$0 \longrightarrow D_S(M'') \xrightarrow{v^t} D_S(M) \xrightarrow{u^t} D_S(M') \longrightarrow 0.$$

- (i) *The homomorphism v^t induces an isomorphism from $D_S(M'')$ onto the kernel of u^t , and u^t is faithfully flat and quasi-compact.*
- (ii) ¹² *$D_S(M')$ represents the quotient sheaf $D_S(M)/D_S(M'')$ for the (fpqc) topology.*

Let $\mathcal{M}$ denote the family of faithfully flat quasi-compact morphisms. First, (ii) follows from (i) (cf. IV, 4.6.5.1). Indeed, the equivalence relation on $D_S(M)$ defined by u^t is the same as that defined by the subgroup

$$\text{Ker}(u^t) = D_S(M'').$$

Since $u^t \in \mathcal{M}$, this equivalence relation is $\mathcal{M}$ -effective (cf. IV, 3.3.2.1); therefore $D_S(M')$ represents the quotient sheaf for the (fpqc) topology (cf. IV, 4.6.5).

The first assertion of (i) is an immediate consequence of the definition of the functors $D_S(-)$. More generally, every exact sequence

$$M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

with no zero on the left gives rise to an exact transposed sequence

$$0 \longrightarrow D_S(M'') \longrightarrow D_S(M) \longrightarrow D_S(M').$$

(This remains valid in the more general setting considered at the beginning of §1.) On the other hand, because $D_S(M)$ and $D_S(M')$ are affine over S , the morphism u^t is necessarily affine, and a fortiori quasi-compact, whatever the homomorphism $u : M' \rightarrow M$. The second assertion of (i) will therefore follow from part (a) of the following corollary.

Corollary 3.2. *Let S be a nonempty prescheme, let $u : M' \rightarrow M$ be a homomorphism of ordinary commutative groups, and let $u^t : G \rightarrow G'$ be the transposed homomorphism. Then:*

¹²Editor's note: what follows has been added, and the proof has been expanded accordingly.

- (a) u is a monomorphism if and only if u^t is faithfully flat;
- (b) u is an epimorphism if and only if u^t is a monomorphism; in that case u^t is in fact a closed immersion.

To prove (a), observe that if u is a monomorphism, then $\mathcal{O}_S(M)$, as a module over $\mathcal{O}_S(M')$, has a nonempty basis: namely, the system of sections defined by any system of representatives of M modulo M' . It is therefore faithfully flat. Conversely, if $\mathcal{O}_S(M)$ is faithfully flat over $\mathcal{O}_S(M')$, then

$$u^t : \mathcal{O}_S(M') \longrightarrow \mathcal{O}_S(M)$$

is injective, which, when $S \neq \emptyset$, implies that $u : M' \rightarrow M$ is injective.

To prove (b), observe that if u is an epimorphism, then $\mathcal{O}_S(M') \rightarrow \mathcal{O}_S(M)$ is surjective, so u^t is a closed immersion and, a fortiori, a monomorphism. Conversely, if u^t is a monomorphism, then $\text{Ker } u^t$ is the unit group. Putting $M'' = \text{Coker } u$, we have seen that

$$\text{Ker } u^t \simeq D_S(M'').$$

Thus 2.1(d) gives $M'' = 0$, and u is an epimorphism.

The usual argument now gives the following consequence of 3.1.

Corollary 3.3. *Let*

$$M' \xrightarrow{u} M \xrightarrow{v} M''$$

be an exact sequence of ordinary commutative groups, and consider the transposed sequence

$$G'' \xrightarrow{v^t} G \xrightarrow{u^t} G'.$$

Then v^t induces a faithfully flat quasi-compact morphism from G'' to $\text{Ker } u^t$, and this kernel is a diagonalizable group isomorphic to

$$D_S(v(M)) = D_S(\text{Coker } u).$$

Corollary 3.4. *Let S be a prescheme, let $u : G \rightarrow H$ be a homomorphism of S -group preschemes that are locally diagonalizable, with H of finite type over S , and put $G' = \text{Ker } u$. Then:*

- (a) G' is locally diagonalizable, and is of finite type over S if G is;
- (b) the quotient G/G' “exists”: more precisely, the equivalence relation on G defined by G' is $\mathcal{M}$ -effective, where $\mathcal{M}$ is the family of faithfully flat quasi-compact morphisms (cf. IV, 3.4). Moreover, G/G' is locally diagonalizable and of finite type over S ;
- (c) the homomorphism $u : G \rightarrow H$ factors uniquely as

$$G \xrightarrow{v} G/G' \xrightarrow{w} H,$$

where v is the canonical homomorphism, and is therefore faithfully flat and quasi-compact. Moreover, w is a closed immersion, and a fortiori a monomorphism.

Finally, the quotient

$$H' = H / \text{Im } w = \text{Coker } w = \text{Coker } u$$

exists: more precisely, the equivalence relation on H defined by G/G' is $\mathcal{M}$ -effective, and H' is of finite type over S .

The first assertion of (c) follows from (b), by the definition of the quotient G/G' (cf. IV, 3.2.3).¹³ Let us show that the (fpqc) quotient sheaf $\widetilde{G}/\widetilde{G'}$ is representable. This can be checked locally on S , as can all the other assertions; we may therefore suppose that G and H are diagonalizable, of the form $D_S(M)$ and $D_S(N)$.

Since H is of finite type over S , the group N is of finite type by 2.1(b). Therefore, by 1.5, u is defined by a homomorphism $u' : N \rightarrow M$. It follows from 3.1 and 3.2 that

$$G' \simeq D_S(\text{Coker } u')$$

and that $\widetilde{G}/\widetilde{G'}$ is represented by $D_S(\text{Im } u')$. Moreover, on considering the exact sequence

$$0 \longrightarrow \text{Ker } u' \longrightarrow N \xrightarrow{u'} \text{Im } u' \longrightarrow 0,$$

we find that w is a closed immersion and that

$$H' = H / \text{Im } w = D_S(\text{Ker } u').$$

The latter is of finite type over S , since N , and hence $\text{Ker } u'$, is of finite type.

Remarks. The existence result for quotients in 3.4 will be substantially generalized in §5.

On the other hand, note that in the present section and the preceding one, the hypothesis $S \neq \emptyset$ was used only to ensure the validity of certain converses, allowing properties of the corresponding ordinary groups to be deduced from certain hypotheses on diagonalizable S -groups. The results in the “forward” direction hold without any restriction on S , and the proofs given here apply in general.

Corollary 3.5. *Let G be a diagonalizable group prescheme over S , and let $n \neq 0$ be an integer. Then the subgroup ${}_nG$ of G , the kernel of the homomorphism*

$$n \cdot \text{id}_G : G \longrightarrow G,$$

is integral over S , and is finite over S if G is of finite type over S .

Indeed, if $G = D_S(M)$, then

$${}_nG = D_S(M/nM)$$

by 3.1, and the result follows from 2.1(b), (c), and (c').

4. Torsors under a Diagonalizable Group

Let S be a prescheme and let $G = D_S(M)$ be a diagonalizable group over S . We shall determine the G -torsors (or principal homogeneous G -bundles) over S for the faithfully flat quasi-compact topology; cf. Exposé IV, 5.1. Recall that a prescheme P over S , with S -operator group G , is called a *torsor* or a *principal homogeneous bundle* if every point of S has an open neighborhood U and a faithfully flat quasi-compact morphism $S' \rightarrow U$ such that

$$P' = P \times_S S'$$

is isomorphic, as an object with operators, to

$$G' = G \times_S S',$$

¹³Editor's note: the original has been expanded in what follows.

where G' acts on itself by right translations. Since G is affine over S , SGA 1, VIII 5.6 implies that P is necessarily affine over S . Also, because G itself is faithfully flat and quasi-compact over S , P is principal homogeneous under G if and only if it is formally principal homogeneous and is, in addition, faithfully flat and quasi-compact over S (cf. IV, 5.1.6).

Recall also (Exposé I, 4.7.3) that giving an S -prescheme P , affine over S , with S -operator group $G = D_S(M)$, amounts to giving a commutative quasi-coherent algebra graded by M over S . In other words, it amounts to giving a quasi-coherent algebra $\mathcal{A}$ over S , together with a direct-sum decomposition as a module

$$\mathcal{A} = \coprod_{m \in M} \mathcal{A}_m,$$

such that

$$\mathcal{A}_m \cdot \mathcal{A}_{m'} \subseteq \mathcal{A}_{m+m'} \quad \text{for } m, m' \in M.$$

With this understood, the answer to the preceding problem is as follows.

Proposition 4.1. *Let P be the prescheme with S -operator group $G = D_S(M)$ defined by the M -graded algebra $\mathcal{A}$. Then P is a principal homogeneous bundle under G if and only if $\mathcal{A}$ satisfies the following conditions:*

- (a) *for every $m \in M$, $\mathcal{A}_m$ is an invertible module over S ;*
- (b) *for $m, m' \in M$, the homomorphism*

$$\mathcal{A}_m \otimes_{\mathcal{O}_S} \mathcal{A}_{m'} \longrightarrow \mathcal{A}_{m+m'}$$

induced by multiplication in $\mathcal{A}$ is an isomorphism.

The necessity of these conditions follows immediately by descent, since they hold when P is the trivial principal homogeneous bundle, that is, when $\mathcal{A} = \mathcal{O}_S(M)$. For sufficiency, note that (a) already implies that P is faithfully flat over S ; it is in any event quasi-compact over S , since it is affine over S . It therefore remains to check that P is formally principal homogeneous under G , that is, that the familiar morphism

$$P \times_S G \longrightarrow P \times_S P$$

is an isomorphism. On affine algebras this morphism is given explicitly by the homomorphism

$$\mathcal{A} \otimes \mathcal{A} \longrightarrow \mathcal{A}(M) = \mathcal{A} \otimes \mathcal{O}_S(M).$$

In bidegree (m, n) , where $m, n \in M$, it is given by

$$x_m \otimes y_n \longmapsto x_m y_n \otimes e_n.$$

At the level of degrees, this homomorphism is compatible with the homomorphism $M \times M \rightarrow M \times M$ given by

$$(m, n) \longmapsto (m + n, n),$$

which is an isomorphism. This shows that (b) expresses precisely, and independently of (a), that P is formally principal homogeneous, and proves 4.1.

Faithfully flat descent also gives:

Corollary 4.2. *The conditions of 4.1 imply that the homomorphism*

$$\mathcal{O}_S \longrightarrow \mathcal{A}_0$$

is an isomorphism.

For example, if $M = \mathbb{Z}$, then, under the conditions of 4.1, the algebra $\mathcal{A}$ is essentially determined by $\mathcal{A}_1 = \mathcal{L}$, namely

$$\mathcal{A} \simeq \prod_{n \in \mathbb{Z}} \mathcal{L}^{\otimes n}$$

as graded algebras. Thus we recover the familiar result:

Corollary 4.3. *The category of principal homogeneous bundles P over S under $\mathbb{G}_{m,S}$ is equivalent to the category of invertible modules $\mathcal{L}$ over S , where the morphisms in both categories are the isomorphisms of the structures in question. Quasi-inverse functors are obtained by assigning to P the degree-1 component of its $\mathbb{Z}$ -graded affine algebra, and by assigning to $\mathcal{L}$ the spectrum of the $\mathbb{Z}$ -graded algebra*

$$\prod_{n \in \mathbb{Z}} \mathcal{L}^{\otimes n}.$$

In particular:

Corollary 4.4. *The group of isomorphism classes of principal homogeneous bundles over S under $\mathbb{G}_{m,S}$ is isomorphic to the group $\text{Pic}(S)$ of isomorphism classes of invertible modules over S , that is, to*

$$H^1(S, \mathcal{O}_S^\times).$$

Since $\mathbb{G}_{m,S}$ is the scheme of automorphisms of the module $\mathcal{O}_S$, Corollary 4.4 is equivalent to the following statement, one of the forms of Hilbert’s “Theorem 90.”

Corollary 4.5. *Every principal homogeneous bundle over S under $\mathbb{G}_m$ is locally trivial for the Zariski topology.*

Remark 4.5.1. The preceding statement is no longer true in general for a group such as μ_n , or for a “twisted form” of $\mathbb{G}_m$. For example, the unique twisted form of $\mathbb{G}_m$ over the field $\mathbb{R}$ of real numbers has a first cohomology group equal to $\mathbb{Z}/2\mathbb{Z}$.

¹⁴ Indeed, let $\mathbb{S}^1$ be the kernel of the norm morphism

$$N : \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}} \longrightarrow \mathbb{G}_{m,\mathbb{R}};$$

it is a $\mathbb{C}/\mathbb{R}$ -twisted form of $\mathbb{G}_{m,\mathbb{R}}$. The equation $N(z) = -1$ in $\prod_{\mathbb{C}/\mathbb{R}} (\mathbb{G}_{m,\mathbb{C}})$ defines an $\mathbb{S}^1$ -torsor X over $\text{Spec}(\mathbb{R})$, locally trivial for the étale topology, but not trivial, since $X(\mathbb{R}) = \emptyset$. Let us show that

$$H_{\text{ét}}^1(\mathbb{R}, \mathbb{S}^1) \simeq \mathbb{Z}/2\mathbb{Z}.$$

There is an exact sequence of smooth commutative algebraic $\mathbb{R}$ -groups

$$1 \longrightarrow \mathbb{S}^1 \longrightarrow \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}} \longrightarrow \mathbb{G}_{m,\mathbb{R}} \longrightarrow 1,$$

which gives rise to a long exact sequence in étale cohomology (or in Galois cohomology):

$$\begin{aligned} 0 \longrightarrow \mathbb{S}^1(\mathbb{R}) \longrightarrow \mathbb{C}^\times \xrightarrow{N} \mathbb{R}^\times \longrightarrow H_{\text{ét}}^1(\mathbb{R}, \mathbb{S}^1) \\ \longrightarrow H_{\text{ét}}^1\left(\mathbb{R}, \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}}\right) \longrightarrow \cdots . \end{aligned}$$

¹⁴Editor’s note: what follows has been added.

Now (see, for example, XXIV, 8.4),

$$H_{\text{ét}}^1\left(\mathbb{R}, \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}}\right) \simeq H_{\text{ét}}^1(\mathbb{C}, \mathbb{G}_{m,\mathbb{C}}),$$

and the latter group is zero by 4.5 (or, in this case, because $\mathbb{C}$ is algebraically closed). We therefore obtain an isomorphism

$$H_{\text{ét}}^1(\mathbb{R}, \mathbb{S}^1) \simeq \mathbb{R}^\times / N(\mathbb{C}^\times) \simeq \{\pm 1\}.$$

We shall need the following result in the next section.

Proposition 4.6. *Under the conditions of 4.1, conditions (a) and (b) are equivalent to the following conditions:*

- (a') *the homomorphism $\mathcal{O}_S \rightarrow \mathcal{A}_0$ is an isomorphism;*
- (b') *for every $m \in M$ (it is enough to require this for the members of a system of generators of M), one has*

$$\mathcal{A}_m \cdot \mathcal{A}_{-m} = \mathcal{A}_0.$$

The necessity is clear in view of 4.2.¹⁵ We are thus reduced to proving the following corollary.

Corollary 4.7. *Let*

$$A = \prod_{n \in \mathbb{Z}} A_n$$

be a $\mathbb{Z}$ -graded ring such that

$$A_1 \cdot A_{-1} = A_0.$$

Then the A_n are invertible A_0 -modules, and, for $n, n' \in \mathbb{Z}$, the homomorphism

$$A_n \otimes_{A_0} A_{n'} \longrightarrow A_{n+n'}$$

induced by multiplication in A is an isomorphism.

By hypothesis there exist $f_i \in A_1$ and $g_i \in A_{-1}$ such that

$$(*) \quad \sum_i f_i g_i = 1.$$

Since the assertion to be proved is local on $\text{Spec}(A_0)$,¹⁶ and since $(*)$ shows that $\text{Spec}(A_0)$ is covered by the affine open subsets $D(f_i g_i)$, we are reduced to the case in which there is an element $f \in A_1$ that is invertible in A . Then, for every $n \in \mathbb{Z}$, f^n is an invertible element of A_n , and therefore defines an isomorphism

$$A_0 \xrightarrow{\sim} A_n, \quad h \longmapsto f^n h.$$

Moreover, this shows that one obtains an isomorphism

$$A_0[t, t^{-1}] \longrightarrow A$$

of graded A_0 -algebras by sending t to f , which completes the proof of 4.7.

¹⁵Editor's note: "4.1" has been corrected to "4.2."

¹⁶Editor's note: the following passage has been expanded.

Under the hypotheses of 4.6, 4.7 already shows that the $\mathcal{A}_m$, for $m \in M$, are invertible. To prove condition 4.1(b), we may therefore suppose that $\mathcal{A}_m$ and $\mathcal{A}_{m'}$ have bases f_m and $f_{m'}$, with inverses

$$f_m^{-1} \in \Gamma(\mathcal{A}_{-m}) \quad \text{and} \quad f_{m'}^{-1} \in \Gamma(\mathcal{A}_{-m'}).$$

Multiplication by

$$f_m^{-1} f_{m'}^{-1} \in \Gamma(\mathcal{A}_{-m-m'})$$

then defines a homomorphism

$$\mathcal{A}_{m+m'} \longrightarrow \mathcal{A}_0 \simeq \mathcal{O}_S$$

that sends the image of $f_m \otimes f_{m'}$ to the unit section of $\mathcal{O}_S$. In the diagram

$$\begin{array}{ccccc} & & w & & \\ & \nearrow & & \searrow & \\ \mathcal{A}_m \otimes \mathcal{A}_{m'} & \xrightarrow{u} & \mathcal{A}_{m+m'} & \xrightarrow{v} & \mathcal{A}_0 \simeq \mathcal{O}_S \end{array}$$

the maps w and v are therefore epimorphisms of invertible sheaves, hence isomorphisms; consequently u is an isomorphism. $\square$

5. Quotient of an Affine Scheme by a Diagonalizable Group Acting Freely

¹⁷ We denote by $\mathcal{M}$ the set of faithfully flat quasi-compact morphisms, and recall that torsors are understood with respect to the (fpqc) topology.

Theorem 5.1. *Let S be a prescheme, M an ordinary commutative group, $G = D_S(M)$ the diagonalizable group over S that it defines, and P an S -prescheme affine over S , on which G acts freely on the right.*

Then the equivalence relation defined by G on P is $\mathcal{M}$ -effective (cf. IV, 3.4); that is, the quotient $X = P/G$ exists and P is a torsor over X under $G_X = D_X(M)$. Moreover, P/G is affine over S . More precisely, if P is defined by the M -graded algebra $\mathcal{A}$, then P/G is isomorphic to $\text{Spec}(\mathcal{A}_0)$, where $\mathcal{A}_0 = \mathcal{A}^G$ is the degree-zero component of $\mathcal{A}$.

Proof. Put $X = \text{Spec}(\mathcal{A}_0)$. The inclusion $\mathcal{A}_0 \rightarrow \mathcal{A}$ gives a natural morphism $P \rightarrow X$, which is invariant under the action of G . Thus P becomes an X -prescheme, affine over X , with operator group $G_X = D_X(M)$; the hypothesis that G acts freely on P/S implies that G_X acts freely on P/X . It remains only to show that P is a principal homogeneous bundle under G_X , using the fact that $\mathcal{B}_0 = \mathcal{O}_X$, where $\mathcal{B}$ is the M -graded algebra over X defining P/X .

We may therefore suppose that $X = S$ and that S is affine. Then P is affine, defined by an M -graded ring A ; write A_m for its homogeneous part of degree m , so that $S = \text{Spec}(A_0)$. In view of 4.6, it remains to verify

$$(x) \quad A_m \cdot A_{-m} = A_0 \quad \text{for every } m \in M.$$

A direct calculation also shows that (x) is equivalent to saying that

$$P \times_S G \longrightarrow P \times_S P$$

¹⁷Editor's note: We have added the following sentence.

is a closed immersion, rather than merely a monomorphism (as follows from the hypothesis that G acts freely); equivalently, the homomorphism of affine rings

$$\theta : A \otimes_{A_0} A \longrightarrow A(M)$$

is surjective.¹⁸ This proves 5.1 if one assumes that the equivalence relation defined by G on P is closed. We shall show, however, that this hypothesis already follows from the fact that G acts freely. This is also implicit in Theorem 5.1: the scheme $G \times_S P$ must be isomorphic to $P \times_X P$, which is closed in $P \times_S P$ because X is affine, hence separated, over S .

Let $R = P \times_S G$. The hypothesis that G acts freely, that is, that $R \rightarrow P \times_S P$ is a monomorphism, says that the diagonal morphism

$$R \longrightarrow R \times_{(P \times_S P)} R = R'$$

is an isomorphism. We have $R = \text{Spec}(A(M))$ and

$$R' = \text{Spec}(A(M \times M)/K),$$

¹⁹ where K is the ideal generated by the elements

$$x_m(e_{m,0} - e_{0,m}), \quad m \in M, \quad x_m \in A_m.$$

Let

$$\phi : A(M \times M) \longrightarrow A(M)$$

be the surjective ring homomorphism defined by

$$xe_{m,n} \longmapsto xe_{m+n} \quad (m, n \in M, x \in A),$$

where the e_m , respectively $e_{m,n} = e_m \otimes e_n$, are the elements of the canonical basis of $A(M)$, respectively $A(M \times M)$. The diagonal morphism $R \rightarrow R'$ then corresponds to the homomorphism

$$\bar{\phi} : A(M \times M)/K \longrightarrow A(M)$$

obtained by passing to the quotient by K . The kernel of $\bar{\phi}$ is the ideal K' generated by

$$d_m = e_{m,0} - e_{0,m}.$$

[The source prints $\ker(\bar{\phi}) = K'$; after quotienting by K , the corresponding kernel is K'/K .] We have $K \subseteq K'$, and the hypothesis that G acts freely on P , that is, that $\bar{\phi}$ is an isomorphism, is equivalent to $K' = K$, expressed by the relations

$$(xx) \quad d_m \in K = \sum_p A(M \times M) A_p d_p \quad \text{for every } m \in M.$$

Using the natural trigrading of $A(M \times M)$, and the fact that the first degree of d_m is zero, this says that d_m can be written as a sum of elements of the form

$$f e_{r,s}(e_{p,0} - e_{0,p}), \quad f \in A_{-p} \cdot A_p.$$

¹⁸Editor's note: Indeed, for every $b \in A$ and $a_m \in A_m$, one has $\theta(b \otimes a_m) = ba_m \otimes e_m$; hence the surjectivity of θ is equivalent to (x).

¹⁹Editor's note: We have denoted by K the ideal denoted by J in the original, in order to distinguish it from the ideals J_p of A_0 occurring in (xxx). We have also made explicit below that the relations (xx) are equivalent to the equality $K = K'$; see what follows.

Since the total degree of d_m is m , it is enough to use terms satisfying

$$r + s + p = m.$$

Thus, for every $m \in M$, there must be an expression

$$(xxx) \quad \begin{cases} d_m = e_{m,0} - e_{0,m} = \sum_{r,s} \lambda_{r,s} (e_{m-s,s} - e_{r,m-r}), \\ \lambda_{r,s} \in J_p = A_p \cdot A_{-p} \subseteq A_0, \quad p = m - (r + s). \end{cases}$$

We must deduce (x), that is, the relations

$$(xxxx) \quad J_n = A_0 \quad \text{for every } n \in M.$$

For this it is enough to establish the same relation modulo every maximal ideal of A_0 . Since all the hypotheses are preserved under such a reduction, we may assume that A_0 is a field.

Lemma 5.2. *Under the preceding conditions, with A_0 a field, if $M \neq 0$, there exists $p \in M - \{0\}$ such that $J_p = A_0$.*

Indeed, otherwise the sum on the right-hand side of (xxx) would vanish for every $m \in M$, which is absurd.

Corollary 5.3. *Under the preceding conditions, but without assuming that A_0 is a field, there are finitely many elements $p_i \in M - \{0\}$ such that*

$$\sum_i J_{p_i} = A_0.$$

Indeed, apply 5.2 to the situations obtained from A/A_0 by reduction modulo the maximal ideals of A_0 .²⁰

Corollary 5.4. *Suppose again that A_0 is a field. For every subgroup N of M with $N \neq M$, there exists $p \in M - N$ such that $J_p = A_0$.*

Indeed, let $M' = M/N$, and consider the M' -graded ring A' , whose underlying ring is A and whose grading is given by

$$A'_{m'} = \coprod_{m \in h^{-1}(m')} A_m,$$

where $h : M \rightarrow M' = M/N$ is the canonical homomorphism. Geometrically, this construction amounts to considering the subgroup $G' = D_S(M')$ of G , and the structure on P of a scheme with operator group G' induced by the action of G . It is then clear that G' acts freely on P [the source prints P']; that is, the pair (M', A') satisfies the hypotheses of 5.3. Thus

$$1 = \sum_i f_i g_i, \quad f_i \in A'_{m'_i}, \quad g_i \in A'_{-m'_i}, \quad m'_i \in M' - \{0\}.$$

Taking the components of the right-hand side along A_0 , and using the fact that A_0 is a field, immediately gives 5.4.

Now observe that

$$J_p J_q \subseteq J_{p+q} \quad \text{and} \quad J_p = J_{-p}.$$

²⁰Editor's note: It follows from 5.2 that $\sum_{p \neq 0} J_p = A_0$; hence 1 can be written as a finite sum $\sum_i x_i$, with $x_i \in J_{p_i}$.

Thus, if N denotes the set of $m \in M$ such that $J_m = A_0$ [the source prints $J_p = A_0$], then N is a subgroup of M . By 5.4, it equals M . This completes the proof of Theorem 5.1.

As noted in the course of the proof, Theorem 5.1 implies:

Corollary 5.5. *Under the hypotheses of 5.1, the graph morphism*

$$P \times_S G \longrightarrow P \times_S P$$

is a closed immersion.

It follows immediately that:

Corollary 5.6. *Let σ be a section of P over S . Then the morphism $g \mapsto \sigma \cdot g$ from G to P defined by σ is a closed immersion.*

Corollary 5.7. *Let G and H be two S -groups, with G diagonalizable and H affine over S , and let $u : G \rightarrow H$ be a homomorphism of S -groups that is a monomorphism. Then u is a closed immersion, $H/G = X$ exists, H is a principal homogeneous bundle over X under G_X , and X is affine over S .*

Corollary 5.8. *Under the hypotheses of 5.1, if P is of finite type (respectively, of finite presentation) over S , then the same is true of $X = P/G$.*

Indeed, the hypothesis implies that the fibers of G_X are of finite type. Hence G_X is of finite presentation over X , by 2.1(b); since P is a torsor under G_X , it is therefore of finite presentation over X .²¹ It is also faithfully flat over X , so the conclusion follows from Exposé V, Proposition 9.1.

6. Essentially Free Morphisms and Representability of Certain Functors of the Form $\Pi_{Y/S} Z/Y^{(*)}$

Definition 6.1. Let $f : X \rightarrow S$ be a morphism of preschemes. We say that f is *essentially free*, or that X is *essentially free over S* , if there are an affine open covering (S_i) of S , for each i an S_i -prescheme S'_i that is affine and faithfully flat over S_i , and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine opens X'_{ij} , such that, for every (i, j) , the ring of X'_{ij} is a free module over the ring of S'_i .²²

Proposition 6.2.

- (a) *If X is essentially free over S , then it is flat over S . The converse holds if S is Artinian.*
- (b) *If S is the spectrum of a field, every S -prescheme is essentially free over S .*
- (c) *If X is essentially free over S , and if $S' \rightarrow S$ is a base-change morphism, then $X' = X \times_S S'$ is essentially free over S' . The converse holds if $S' \rightarrow S$ is faithfully flat and quasi-compact.*

²¹Editor's note: Cf. EGA II, 2.7.1(vi).

(*) This section is independent of the theory of diagonalizable groups; its natural place would be in Exposé VI_B.

²²Editor's note: "Free" may be replaced by "projective"; cf. 6.8 below. This notion should also be compared with that of an S -prescheme that is flat and pure, introduced and developed in [RG71]; see in particular *loc. cit.*, Part I, 3.3.12, and Part II, 3.1.4.1.

The proof is immediate, using for the converse in (a) the fact that a flat module over an Artinian local ring is free.²³

Proposition 6.3. *Let H be a diagonalizable S -group prescheme (more generally, one that becomes diagonalizable after a suitable faithfully flat quasi-compact extension of every affine open of S ; that is, H is “of multiplicative type,” cf. IX, 1.1). Then H is essentially free over S .*

Indeed, if H is diagonalizable, it is affine over S and is defined by an algebra that is a free $\mathcal{O}_S$ -module.

The introduction of Definition 6.1 is justified by the following theorem.

Theorem 6.4. *Let S be a prescheme, Z an S -prescheme essentially free over S , and Y a closed subprescheme of Z . Consider the functor²⁴*

$$F = \prod_{Z/S} Y/Z : (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Set},$$

$$F(S') = \Gamma(Y_{S'}/Z_{S'}) = \begin{cases} \emptyset, & Z_{S'} \neq Y_{S'}, \\ \{\mathrm{id}_{Z_{S'}}\}, & Z_{S'} = Y_{S'}. \end{cases}$$

This functor is representable by a closed subprescheme of S .²⁵

²⁶ First note that F is a sheaf for the (fpqc) topology. Since $F(S')$ is either $\emptyset$ or $\{\mathrm{pt}\}$ for every S' , it is enough to check the following assertion: if (S_i) is an open covering of S (respectively, if $S' \rightarrow S$ is faithfully flat and quasi-compact), and every $Y_{S_i} \rightarrow Z_{S_i}$ is an isomorphism (respectively, $Y_{S'} \rightarrow Z_{S'}$ is an isomorphism), then $Y \rightarrow Z$ is an isomorphism. This is clear in the first case and follows from SGA 1, VIII, 5.4, or EGA IV₂, 2.7.1, in the second.

Moreover, by SGA 1, VIII, 2.1 and 5.5, faithfully flat quasi-compact morphisms are morphisms of effective descent for the fibered category of arrows that are closed immersions. Thus, with the notation of 6.1, we may reduce to the case $S = S'_i$.

Let (Z_j) be a covering of Z by affine opens such that $\mathcal{O}(Z_j)$ is a free module over $A = \mathcal{O}(S)$, put $Y_j = Y \cap Z_j$, and let

$$F_j : (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Set}$$

be the functor defined from (Z_j, Y_j) as F is defined from (Z, Y) . It is a subfunctor of the terminal functor, and clearly

$$F = \bigcap_j F_j.$$

It therefore suffices to prove that every F_j is representable by a closed subscheme T_j of S , for then F is represented by the closed subscheme $T = \bigcap_j T_j$. We may consequently suppose that Z is affine as well, say $Z = \mathrm{Spec}(B)$, where B is a free A -module. Let J be a subset of B defining the closed subprescheme Y of Z , and let K be the ideal of A

²³Editor's note: Indeed, let $(A, \mathfrak{m})$ be an Artinian local ring, k its residue field, M an arbitrary A -module, and $(x_i)_{i \in I}$ elements of M whose images form a basis of $M/\mathfrak{m}M$ over k . Let F be the free A -module with basis $(e_i)_{i \in I}$, and let $\phi : F \rightarrow M$ be the A -morphism defined by $\phi(e_i) = x_i$. Then $Q = \mathrm{Coker} \phi$ satisfies $Q = \mathfrak{m}Q$, whence $Q = 0$, since $\mathfrak{m}$ is nilpotent. Suppose moreover that M is flat over A . Then $K = \mathrm{Ker} \phi$ satisfies $K \otimes_A k = 0$, that is, $K = \mathfrak{m}K$, whence $K = 0$.

²⁴Editor's note: Cf. II.1, where this functor is denoted $\prod_{Z/S} Y$.

²⁵Editor's note: We have corrected Z to S .

²⁶Editor's note: We have expanded the original in what follows; see also 1.7.3.

generated by the $u_i(J) \subseteq A$, where the $u_i : B \rightarrow A$ are the coordinate forms for the chosen basis. It is immediate that

$$T = \mathcal{V}(K) = \operatorname{Spec}(A/K)$$

has the required property. This completes the proof.

Examples 6.5. We give some important examples of functors that reduce to functors $\prod_{Z/S} Y/Z$ of the type considered in 6.4, and for which criteria of representability will be useful below. Here S denotes a prescheme, and $X, Y, Z, \dots$ denote preschemes over S .

(a) Suppose given an S -morphism

$$(x) \quad q : X \longrightarrow \underline{\operatorname{Hom}}_S(Y, Z),$$

(“ X acts on Y with values in Z ”), that is, a morphism

$$(xx) \quad r : X \times_S Y \longrightarrow Z.$$

Let Z' be a subprescheme of Z . It gives a monomorphism

$$\underline{\operatorname{Hom}}_S(Y, Z') \longrightarrow \underline{\operatorname{Hom}}_S(Y, Z),$$

making the first functor a subfunctor of the second. Let X' be the inverse image of this subfunctor under (x). Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Z_T$$

factors through Z'_T . This functor has the following description. Put $P = X \times_S Y$, and let P' be the inverse image of Z' under $r : P \rightarrow Z$. Then there is an evident isomorphism

$$(xxx) \quad X' \simeq \prod_{P/X} P'/P.$$

It follows that *if Y is essentially free over S and Z' is closed in Z , then the subfunctor X' of X is representable by a closed subprescheme of X .*

(b) Suppose given two actions of X on Y with values in Z , that is, two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\operatorname{Hom}}_S(Y, Z),$$

and put $X' = \operatorname{Ker}(q_1, q_2)$. This is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms

$$q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$$

are equal. Giving q_1, q_2 is equivalent to giving a morphism

$$q : X \longrightarrow \underline{\operatorname{Hom}}_S(Y, Z \times_S Z),$$

or, equivalently, a morphism

$$r : X \times_S Y \longrightarrow Z \times_S Z.$$

Put $U = Z \times_S Z$, and let U' be the diagonal subprescheme of $Z \times_S Z$. Then X' is the inverse image under q of the subfunctor

$$\underline{\operatorname{Hom}}_S(Y, U') \longrightarrow \underline{\operatorname{Hom}}_S(Y, U).$$

It can therefore be written in the form (xxx), with $P = X \times_S Y$ and P' the inverse image of the diagonal under r , that is, the kernel of

$$X \times_S Y \xrightarrow[r_2]{r_1} Z.$$

Thus we are in the situation of (a).

Consequently, *if Y is essentially free over S and Z is separated over S , then the subfunctor X' of X is representable by a closed subprescheme of X .*

(c) Suppose given a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y);$$

that is, “ X acts on Y .” Let X' be the “kernel” of this morphism, namely the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \rightarrow Y_T$ is the identity. This functor falls under (b), as one sees by introducing a second morphism

$$q' : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y)$$

that “makes X act trivially on Y .” Hence, *if Y is essentially free and separated over S , then the kernel subfunctor of q is representable by a closed subprescheme of X .*

(d) Under the hypotheses of (c), consider the subfunctor Y' of Y of “invariants under X .” Thus $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $\bar{q}(y) : X_T \rightarrow Y_T$ is the “ T -constant morphism with value y .” Introducing q' as in (c), and writing $\bar{q}$ and $\bar{q}'$ for the corresponding morphisms,

$$\bar{q}, \bar{q}' : Y \rightrightarrows \underline{\mathrm{Hom}}_S(X, Y),$$

one sees that $Y' = \mathrm{Ker}(\bar{q}, \bar{q}')$. It therefore again falls under (b), with the roles of X and Y reversed and $Z = Y$. Consequently, *if X is essentially free over S and Y is separated over S , then the subfunctor Y' of Y of invariants under X is representable by a closed subprescheme of Y .*

(e) Constructions of the type described in the preceding examples occur especially often in group theory. Thus, when a group prescheme G over S acts on the S -prescheme X ,

$$q : G \longrightarrow \underline{\mathrm{Aut}}_S(X),$$

the kernel of q , the “subgroup of G acting trivially,” is a closed subprescheme of G provided that X is essentially free and separated over S , by (c). The subobject X^G of invariants is a closed subprescheme of X , provided that G is essentially free over S and X is separated over S ,²⁷ by (d).

Let Y, Z be subpreschemes of X , and consider the subfunctor $\underline{\mathrm{Transp}}_G(Y, Z)$ of G , the “transporter of Y into Z .” Its points with values in an S -prescheme T are the $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subseteq Z_T$, that is, induces a morphism $Y_T \rightarrow Z_T$ factoring through Z_T . Hence, *if Y is essentially free over S , and Z is closed in X , then $\underline{\mathrm{Transp}}_G(Y, Z)$ is a closed subprescheme of G , by (a).*

One may also consider the *strict transporter* of Y into Z ,²⁸ whose T -valued points are the $g \in G(T)$ such that $g(Y_T) = Z_T$. It is precisely

$$\underline{\mathrm{Transp}}_G(Y, Z) \cap \sigma(\underline{\mathrm{Transp}}_G(Z, Y)),$$

²⁷Editor’s note: We have corrected S_X to S .

²⁸Editor’s note: It is denoted $\underline{\mathrm{Transp}}_G^{\mathrm{str}}(Y, Z)$.

where σ is the symmetry of G . Consequently, if Y and Z are essentially free over S and closed in X , then the strict transporter of Y into Z is a closed subprescheme of G .

An important case is $X = G$, with G acting on itself by inner automorphisms. If H is a subprescheme of G , the strict transporter of H into itself is also called the *normalizer of H in G* , and is denoted $\underline{\text{Norm}}_G H$. Hence, if H is a closed subgroup prescheme of G , essentially free over S , then $\underline{\text{Norm}}_G H$ is representable by a closed subgroup prescheme of G .

Finally, let Z be a subprescheme of G . Its *centralizer* $\underline{\text{Centr}}_G(Z)$ in G is the group subfunctor of G defined by the procedure in (d), regarding “ Z acts on G ” through the action induced by that of G . Thus, if Z is essentially free over S and G is separated over S , then $\underline{\text{Centr}}_G(Z)$ is a closed subgroup prescheme of G . In particular, if G is essentially free and separated over S , then the center C of G , which is none other than $\underline{\text{Centr}}_G(G)$, is a closed subgroup prescheme of G .

When S is the spectrum of a field, 6.2(b) [the source prints 6.3(b)] shows that in examples (a)–(e) above the “essentially free” hypotheses are automatically satisfied; only separation hypotheses remain. Recalling that a group prescheme over a field is necessarily separated, one obtains, for example:

Corollary 6.7. ²⁹ Let G be a group prescheme over a field k . Then:

- For every subprescheme Z of G , the centralizer of Z in G is a closed subgroup prescheme of G . In particular, this holds for the center $\underline{\text{Centr}}_G(G)$ of G .
- More generally, if $u, v : X \rightarrow G$ are morphisms of preschemes, then $\underline{\text{Transp}}_G(u, v)$ is representable by a closed subprescheme of G .
- For two subpreschemes Y, Z of G , with Z closed, $\underline{\text{Transp}}_G(Y, Z)$ is a closed subprescheme of G . If Y is also closed, the same conclusion holds for $\underline{\text{Transp}}_G^{\text{str}}(Y, Z)$.
- For every subgroup prescheme³⁰ H of G , the normalizer $\underline{\text{Norm}}_G(H)$ is a closed subgroup prescheme of G .

Remark 6.8. ³¹ Let A be a commutative ring, let M be an A -module, and put

$$M^\vee = \text{Hom}_A(M, A).$$

Endow the A -module $\text{End}_A(M)$ with the topology of discrete pointwise convergence. Thus a basis of neighborhoods of zero is formed by the following A -submodules, where $n \in \mathbb{N}$ and $m_1, \dots, m_n \in M$:

$$K(m_1, \dots, m_n) = \{u \in \text{End}_A(M) \mid u(m_i) = 0 \text{ for } i = 1, \dots, n\}.$$

We say that M is a *quasi-free A -module* if the image of the canonical morphism

$$\Theta : M \otimes_A M^\vee \longrightarrow \text{End}_A(M)$$

contains id_M in its closure; that is, if the following condition holds:

²⁹Editor’s note: We have retained the numbering of the original; there is no 6.6.

³⁰Editor’s note: Indeed, over a field k , every subgroup prescheme of G is closed; cf. Exposé VI_A, 0.6.1.

³¹Editor’s note: We have expanded the original in what follows; in particular, we have added Lemma 6.8.1.

$$(*) \quad \begin{cases} \text{for all } m_1, \dots, m_n \in M, \text{ there exist } x_1, \dots, x_r \in M \text{ and } f_1, \dots, f_r \in M^\vee \\ \text{such that } m_i = \Theta \left(\sum_{s=1}^r x_s \otimes f_s \right) (m_i) = \sum_{s=1}^r f_s(m_i) x_s \quad \text{for } i = 1, \dots, n. \end{cases}$$

(In this case $\text{Im } \Theta$ is dense in $\text{End}_A(M)$, since for every $u \in \text{End}_A(M)$ one has $u(m_i) = \sum_{s=1}^r f_s(m_i) u(x_s) = \Theta(\sum_{s=1}^r u(x_s) \otimes f_s)(m_i)$.)

First note that this property is stable under base change. Indeed, let $\phi : A \rightarrow A'$ be a ring homomorphism, put $M' = M \otimes_A A'$, and let $m'_1, \dots, m'_n \in M'$. Write

$$m'_i = \sum_j m_{ij} \otimes b_{ij}, \quad m_{ij} \in M, \quad b_{ij} \in A'.$$

By hypothesis there exist $x_1, \dots, x_r \in M$ and $f_1, \dots, f_r \in \text{Hom}_A(M, A)$ such that

$$m_{ij} = \sum_s x_s f_s(m_{ij}) \quad \text{for all } i, j.$$

Write $\phi \circ f_s$ for the image of f_s in

$$\text{Hom}_A(M, A') = \text{Hom}_{A'}(M', A').$$

Then, for $i = 1, \dots, n$,

$$\begin{aligned} \left(\sum_s x_s \otimes \phi \circ f_s \right) (m'_i) &= \sum_{s,j} x_s \otimes \phi(f_s(m_{ij})) b_{ij} \\ &= \sum_j \left(\sum_s x_s f_s(m_{ij}) \right) \otimes b_{ij} \\ &= m'_i. \end{aligned}$$

This proves that M' is quasi-free over A' .

Also, every projective A -module P is quasi-free. Indeed, there are A -morphisms

$$A^{(I)} \xrightarrow{\pi} P \xrightarrow{\tau} A^{(I)} \quad \text{such that} \quad \pi \circ \tau = \text{id}_P.$$

Let (e_i) be the canonical basis of $A^{(I)}$, and put $f_i = e_i^* \circ \tau$, a linear form on P . Given $m_1, \dots, m_n \in P$, there is a finite subset J of I such that

$$m_k = \sum_{i \in J} f_i(m_k) \pi(e_i) \quad \text{for } k = 1, \dots, n.$$

Theorem 6.4 remains valid if, in the statement of Definition 6.1, “free” is replaced by “projective” or, more generally, by “quasi-free.” Proceeding as in the proof of 6.4, one reduces to the following lemma.

Lemma 6.8.1. *Let M be a quasi-free A -module, N a submodule, and F the covariant functor*

$$F : (A\text{-algebras}) \longrightarrow \mathbf{Set}$$

such that $F(B) = \{\text{pt}\}$ if $M_B = (M/N)_B$, and $F(B) = \emptyset$ otherwise. Then there is an ideal K of A such that $F(B) = \{\text{pt}\}$ if and only if the morphism $A \rightarrow B$ factors through A/K ; that is, there is an isomorphism, functorial in B ,

$$F(B) \simeq \text{Hom}_{A\text{-alg.}}(A/K, B).$$

Proof. Let (n_j) be a system of generators of N , and let F_j be the subfunctor of the terminal functor e , where $e(B) = \{\text{pt}\}$ for every B , corresponding to the submodule An_j . We have $F(B) = \{\text{pt}\}$ if and only if the image of every n_j in M_B is zero. Hence F is the intersection of the functors F_j , reducing us to the case in which N is generated by one element n .

Let B be an A -algebra, with structure map $\phi : A \rightarrow B$. If the image $n \otimes 1$ of n in M_B is zero, then for every $f \in M^\vee = \text{Hom}_A(M, A)$,

$$0 = f(n) \otimes 1 = \phi(f(n)).$$

Conversely, because M is quasi-free, there exist $x_1, \dots, x_r \in M$ and $f_1, \dots, f_r \in M^\vee$ such that

$$n = \sum_s f_s(n) x_s,$$

whence

$$n \otimes 1 = \sum_s x_s \otimes \phi(f_s(n)).$$

It follows that $n \otimes 1 = 0$ if and only if ϕ factors through A/K , where K is the ideal generated by the $f_s(n)$, for $s = 1, \dots, r$. This proves the lemma.

7. Appendix: On Monomorphisms of Group Preschemes

The result proved in this section is not needed in the remainder of the seminar, except in X, 8.8 and Exposés XV and XVI. It depends essentially on the existence theorem for quotient groups in Exposé VI_A.³²

Corollary 5.7 raises the question of when a monomorphism $u : G \rightarrow H$ of S -groups can be asserted to be an immersion, or even a closed immersion.

We saw in VI_B, 1.4.2 that this is so when S is the spectrum of a field k , provided that G is of finite type over k and H is locally of finite type over k . It follows readily that the same result remains valid when S is merely assumed Artinian.³³

On the other hand, examples of bijective monomorphisms that are not immersions are easy to give, for instance over the affine line over a field or over the spectrum of a discrete valuation ring. Take

$$H = (\mathbf{Z}/2\mathbf{Z})_S, \quad G = G_1 \times_S G_2,$$

where G_1 is the open subgroup of H complementary to the closed point $x \neq 0$ in the fiber $H_s \simeq \mathbf{Z}/2\mathbf{Z}$, with s a fixed closed point of S , and G_2 is the closed subscheme of H which is the union of the subscheme given by the identity section and the reduced closed subscheme defined by x . One checks immediately that G_2 is stable under multiplication in H , and hence is a group scheme. The immersions $G_i \rightarrow H$, $i = 1, 2$, then define a homomorphism of S -groups

$$G = G_1 \times_S G_2 \longrightarrow H$$

which is visibly a bijective monomorphism, and even a local immersion, but is not an immersion. Indeed, G and H are reduced; G has three disjoint irreducible components, whereas H has only two.

Notice also that $G_1 \rightarrow H$ is an example of an open subgroup of H that is not closed, in contrast to the situation for algebraic groups over a field. The theory of degenerations

³²Editor's note: cf. Exposé VI_A, Theorems 3.2 and 3.3.2.

³³Editor's note: the additions made in Exposé VI_B should be taken into account.

of elliptic curves gives further examples of this phenomenon in which H is smooth over S of relative dimension one and G has connected fibers.

One might hope^(*) that once G is assumed flat over S , and, say, G and H are of finite presentation over S , every monomorphism $u : G \rightarrow H$ of S -groups is automatically an immersion. We shall prove a result of this kind under additional hypotheses.

First, we may assume that S is affine and, by the finite-presentation hypothesis on G and H , that S is Noetherian: one reduces to the spectrum of a ring of finite type over $\mathbf{Z}$.³⁴ Then G and H are Noetherian. Saying that $u : G \rightarrow H$ is a closed immersion (respectively, an immersion) is then equivalent to saying that u is a monomorphism—which holds by hypothesis—and is proper (respectively, proper at every point of $u(G)$).³⁵

The valuative criterion for properness shows that it is enough to check, after every base change $S' \rightarrow S$ with S' the spectrum of a discrete valuation ring, complete if desired, that the morphism $u' : G' \rightarrow H'$ has the corresponding properness property. The case of local properness was omitted from EGA II, 7.3 and was to appear as an erratum in EGA IV.^(*) We are therefore reduced to the case in which S itself is the spectrum of a complete discrete valuation ring, subject to the proviso that any additional hypotheses imposed on G, H, u be stable under base change.

Let s , respectively s' , be the closed, respectively generic, point of S . The induced homomorphisms on the fibers,

$$u_s : G_s \longrightarrow H_s, \quad u_{s'} : G_{s'} \longrightarrow H_{s'},$$

are closed immersions because they are monomorphisms of group schemes of finite type over fields (VI_B, 1.4.2). We may therefore identify $G_{s'}$ with a closed subscheme of $H_{s'}$. We shall use the following result.

Lemma 7.1. *Let S be the spectrum of a discrete valuation ring, with generic point s' . Let H be an S -prescheme and let L' be a closed subprescheme of the generic fiber $H_{s'}$, hence also a subprescheme of H . Its schematic closure L in H —the smallest closed subprescheme of H containing L' , cf. EGA I, 9.5—exists. It is also the unique closed subprescheme of H , flat over S , whose generic fiber is L' . Formation of L is functorial in the pair (H, L') and commutes with fiber products over S . In particular, if H is an S -group and L' is a subgroup of $H_{s'}$, then L is a subgroup of H .*

The proof is immediate and is left to the reader.^(**) Apply this to the pair $(H, G_{s'})$. The monomorphism $u : G \rightarrow H$ factors as

$$G \xrightarrow{u'} L \longrightarrow H,$$

where L is a subgroup of H , a closed subprescheme flat over S , and $u' : G \rightarrow L$ induces an isomorphism on generic fibers. Thus $u : G \rightarrow H$ is an immersion (respectively, a closed immersion) if and only if $u' : G \rightarrow L$ is one. We are reduced to the case in which H is flat over S and $u_{s'} : G_{s'} \rightarrow H_{s'}$ is an isomorphism, again subject to the requirement that any additional hypotheses remain valid when H is replaced by a closed subgroup prescheme.

Since H is then the schematic closure of $H_{s'}$, if u is an immersion its image $u(G)$ is a subprescheme³⁶ of H containing $H_{s'}$. Its schematic closure is therefore H , so $u(G)$ is an

^(*)In fact, M. Raynaud constructed a counterexample with G smooth and with connected fibers; cf. XVI, 1.1(c). Without the connected fiber hypothesis, one may take $S = \text{Spec}(\mathbf{Z}_2)$, let $G = (\mathbf{Z}/2\mathbf{Z})_S$ with its nonidentity closed point removed, and let $H = (\mu_2)_S$.

³⁴Editor's note: cf. EGA IV₃, §8 and Exposé VI_B, §10.

³⁵Editor's note: cf. EGA IV₃, 8.11.5 and 15.7.1.

^(*)Cf. EGA IV₃, 15.7.

^(**)Cf. EGA IV₂, 2.8.

³⁶Editor's note: the word "closed" has been removed.

open subprescheme of H . We must consequently prove that u is an open immersion, or an isomorphism if the desired conclusion is that it is a closed immersion.

Because G and H are flat over S , this is equivalent to the induced maps on the fibers being open immersions, respectively isomorphisms (cf. SGA 1, I, 5.7). It already holds for $u_{s'}$, so it remains to prove that u_s is an open immersion, respectively an isomorphism.

Since G and H are flat over S , their fiber dimensions are constant (VI_B, 4.3). Their generic fibers agree, hence these dimensions are the same for G and H . Thus $u_s : G_s \rightarrow H_s$ is a monomorphism of algebraic groups of the same dimension. It follows that G_s is open in H_s , and set-theoretically is a union of connected components of H_s . Indeed, after passing to an algebraically closed base field and reducing G and H , both are smooth over the field and the assertion is immediate.

The complement $H_s - G_s$ is closed in H_s , hence in H . Its complement H' in H is open and is stable under the group law. Upon replacing H by H' , we may therefore reduce, with the usual proviso, to the case in which u is bijective, that is, $u_s : G_s \rightarrow H_s$ is bijective. In every case we are reduced to proving that u , equivalently u_s , is an isomorphism, possibly under this bijectivity hypothesis.

Assume first that u is bijective. If H_s is reduced, then u_s is an isomorphism, because G_s identifies with a closed subscheme of H_s having the same underlying set. In particular, if $k = \kappa(s)$ has characteristic zero, every algebraic group over k is reduced by Cartier's theorem (cf. VI_B, 1.6.1; VII_B, 3.3.1; or EGA IV₄, 16.12.2 and 17.12.5). We have proved:

Proposition 7.2. *Let $u : G \rightarrow H$ be a homomorphism of group preschemes of finite presentation over S . If u is a monomorphism, G is flat over S , and every residue field of S has characteristic zero, then u is an immersion.*

Suppose now that $\kappa(s)$ has characteristic $p > 0$, and restrict to the case in which G is commutative. Under the preceding reductions, H is also commutative, because it is the schematic closure of $H_{s'} \simeq G_{s'}$. For $n \geq 0$, set

$$S_n = \operatorname{Spec}(V/\mathfrak{m}^{n+1}), \quad G_n = G \times_S S_n, \quad H_n = H \times_S S_n,$$

where V is the valuation ring defining S and $\mathfrak{m}$ its maximal ideal. For $m > 0$, denote by ${}_mG$ and ${}_mH$ the kernels of multiplication by m in G and H . Define similarly

$${}_m(G_n) = ({}_mG)_n,$$

which we abbreviate to ${}_mG_n$, and likewise for H .

By VI_A, 3.2, the quotients

$$Q_n = H_n/G_n$$

exist. Each Q_n is a commutative group scheme over S_n , flat over S_n , and $H_n \rightarrow Q_n$ is faithfully flat with kernel G_n . Formation of the quotient commutes with base extension,³⁷ so

$$Q_n \simeq Q_m \times_{S_m} S_n \quad (m \geq n).$$

In particular, $Q_n \times_{S_n} S_0 = Q_0 = H_0/G_0$. Since G_0 and H_0 have the same underlying set, Q_0 is set-theoretically a single point: it is a purely infinitesimal group.

It follows that every Q_n is finite and flat over S_n . Let it be defined by an algebra C_n over $V_n = V/\mathfrak{m}^{n+1}$, finite free as a V_n -module. For $m \geq n$,

$$C_n = C_m \otimes_{V_m} V_n,$$

³⁷Editor's note: this should be made explicit in Exposés V and VI_A.

compatibly also with the comultiplication maps. Hence

$$C = \varprojlim C_n \quad \text{is a finite free module over} \quad V = \varprojlim V_n,$$

and the comultiplications on the C_n induce one on C . Thus $Q = \operatorname{Spec}(C)$ is a finite flat group scheme over S satisfying

$$Q \times_S S_n = Q_n$$

for every n .

Lemma 7.3. ³⁸ *Let K be a field and let Q be a finite group scheme over K , of degree n . Then the n -th power morphism on Q is zero.*

See VII_A, 8.5.

Remark 7.3.1. The statement of Lemma 7.3 still makes sense for a finite locally free group scheme Q/S over an arbitrary base prescheme S . It would be interesting to find a proof in that generality.

Lemma 7.3 (that is, VII_A, 8.5) does show that the proposed statement holds whenever S is reduced: apply it to the fibers of Q at the maximal points of S , equivalently the generic points of its irreducible components.³⁹

In the preceding situation, where S is the spectrum of a discrete valuation ring and Q is commutative, it follows that $n \operatorname{id}_Q = 0$. Here n is a power p^{ν_0} of the residue characteristic. Thus:

Corollary 7.4. *With Q and the Q_n as above, suppose that their common degree is p^{ν_0} . Then*

$$p^{\nu_0} \operatorname{id}_Q = 0, \quad p^{\nu_0} \operatorname{id}_{Q_n} = 0 \quad \text{for every } n.$$

Corollary 7.5. *Suppose in addition that G_0 is smooth over k and that $p \operatorname{id}_{G_0}$ is flat. Equivalently, by the structure theory of algebraic groups over an algebraically closed field $\bar{k}$, $G_{\bar{k}}$ contains no subgroup isomorphic to the additive group. Then, for every $\nu \geq \nu_0$ and every $n > 0$, the group $p^{\nu} H_n$ is flat over S_n .*

Indeed, $p^{\nu_0} \operatorname{id}_{Q_n} = 0$ implies that $p^{\nu} \operatorname{id}_{H_n}$ factors through G_n , giving the commutative diagram

$$\begin{array}{ccc} H_n & \xrightarrow{p^{\nu} \operatorname{id}_{H_n}} & H_n \\ \uparrow u_n & \searrow v_n & \uparrow u_n \\ G_n & \xrightarrow{p^{\nu} \operatorname{id}_{G_n}} & G_n. \end{array}$$

We claim that v_n is flat. Since H_n and G_n are flat over S_n , it is enough to check that v_0 is flat (SGA 1, IV, 5.9). Thus we may take $n = 0$, so that $S_n = \operatorname{Spec}(k)$. Since $p \operatorname{id}_{G_0}$, and hence $p^{\nu} \operatorname{id}_{G_0}$, is flat, its image is an induced open subgroup G'_0 of G_0 . The map $u_0 : G_0 \rightarrow H_0$ is surjective, so v_0 takes its values in G'_0 and may be regarded as a homomorphism to G'_0 . Its composite with u_0 is an epimorphism; hence v_0 itself is an epimorphism, and therefore a flat homomorphism into G'_0 , and thus into G_0 . Consequently v_n is flat and $\ker v_n = p^{\nu} H_n$ [†] is flat over S_n . $\square$

³⁸Editor's note: the hypothesis that Q be commutative has been removed here, and 7.3.1 modified accordingly. If Q is not commutative, the n -th power morphism need not be a group homomorphism.

³⁹Editor's note: this should be added to Exposé VII_A. On the other hand, the statement is true for every S when Q is commutative; cf. Deligne's theorem in [TO70], p. 4.

[†]English editor's note: the source prints $p^{\nu} H$ over S here and $p^{\nu} H$ over S_n in the proof of Lemma 7.7. The subscripts n are supplied in both places because those arguments concern the base-changed family over S_n .

Remark. We did not explicitly use the assumption that G_0 is smooth over k . It is, however, an easy consequence of the flatness of pid_{G_0} [the source prints pid_G]; this is why the condition was stated explicitly in Corollary 7.5.

Lemma 7.6. *Let $u : G \rightarrow H$ be a surjective monomorphism of commutative algebraic groups over a field k of characteristic $p > 0$, and put $Q = H/G$, a purely infinitesimal group. There is an integer ν_1 such that, for every $\nu \geq \nu_1$, the sequence*

$$0 \longrightarrow {}_{p^\nu}G \longrightarrow {}_{p^\nu}H \longrightarrow Q \longrightarrow 0$$

is exact.

It is enough to establish exactness at Q , which in turn amounts to the injectivity of the homomorphism

$$(x) \quad \mathcal{O}_{Q,e} \longrightarrow \mathcal{O}_{{}_{p^\nu}H,e}$$

of local rings at the identity elements. Recall that Q is set-theoretically supported at the single point e . There is a natural homomorphism

$$(xx) \quad \mathcal{O}_{{}_{p^\nu}H,e} \longrightarrow \mathcal{O}_{H,e}/\mathfrak{m}^{p^\nu},$$

where $\mathfrak{m}$ is the augmentation ideal, equivalently the maximal ideal, of $\mathcal{O}_{H,e}$. Indeed, $p^\nu \mathrm{id}_H$ vanishes on the kernel of the iterated Frobenius homomorphism F^ν . The composite of (x) and (xx) is the natural composite

$$(xxx) \quad \mathcal{O}_{Q,e} \longrightarrow \mathcal{O}_{H,e} \longrightarrow \mathcal{O}_{H,e}/\mathfrak{m}^{p^\nu}.$$

Now $\mathcal{O}_{Q,e} \rightarrow \mathcal{O}_{H,e}$ is injective, because $H \rightarrow Q$ is an epimorphism and therefore flat. Moreover, $\mathcal{O}_{Q,e}$ is Artinian, while $\bigcap_\nu \mathfrak{m}^{p^\nu} = 0$ in $\mathcal{O}_{H,e}$, so the intersection of the corresponding ideals in $\mathcal{O}_{Q,e}$ is also zero. One of them is therefore zero. Thus (xxx) is injective, and a fortiori so is (x).

Lemma 7.7. *Under the hypotheses of 7.5, there is an integer ν_1 such that, for every $\nu \geq \nu_1$ and every n , the sequence of S_n -groups*

$$0 \longrightarrow {}_{p^\nu}G_n \longrightarrow {}_{p^\nu}H_n \xrightarrow{w_n} Q_n \longrightarrow 0$$

is exact. More precisely, w_n is faithfully flat and its kernel is ${}_{p^\nu}G_n$.

Take ν_1 as in Lemma 7.6, applied to $u_0 : G_0 \rightarrow H_0$, and require $\nu_1 \geq \nu_0$, where $p^{\nu_0} = \mathrm{rank} Q_0$. It remains only to check that w_n is faithfully flat. By Corollary 7.5, ${}_{p^\nu}H_n$ is flat over S_n , as is Q_n ; it is therefore enough to check that

$$w_0 : {}_{p^\nu}H_0 \longrightarrow Q_0$$

is faithfully flat, equivalently an epimorphism. This follows from the choice of ν_1 .

Corollary 7.8. *Suppose in addition that H is separated over S , or more generally that ${}_{p^\nu}H$ is separated over S for every ν , and that ${}_{p^\nu}G$ is finite over S for every ν . Then, for $\nu \geq \nu_1$, the group schemes ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are finite and flat over S , and there is an exact sequence*

$$0 \longrightarrow {}_{p^\nu}G \longrightarrow {}_{p^\nu}H \xrightarrow{w} Q \longrightarrow 0.$$

Because $u : G \rightarrow H$ is surjective, the induced morphism ${}_{p^\nu}G \rightarrow {}_{p^\nu}H$ is also surjective. The source is finite over S and the target is separated over S , hence the target is finite over S . To prove that ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are flat over S , it is enough to prove this after every base

change to S_n , where it is contained in Corollary 7.5. Finally, the exact sequence above is obtained from the exact sequences in Lemma 7.7 as n varies.

Taking generic fibers in the sequence of Corollary 7.8, and recalling that $u_{s'} : G_{s'} \rightarrow H_{s'}$ is an isomorphism, gives that $Q_{s'}$ is the unit group. Since Q is flat over S , Q itself is the unit group; therefore so is Q_s , and $u_s : G_s \rightarrow H_s$ is an isomorphism. We have proved:

Proposition 7.9. *Let $u : G \rightarrow H$ be a homomorphism of group preschemes of finite presentation over a prescheme S . Assume:*

- (a) *u is a monomorphism;*
- (b) *G is flat over S ;*
- (c) *for every $s \in S$ whose residue field has characteristic $p > 0$, let $S' = \operatorname{Spec}(\mathcal{O}_{S,s})$, and let $u' : G' \rightarrow H'$ be obtained from u by base change to S' . Then G' is commutative, the special fiber $G'_0 = G_s$ is smooth over $\kappa(s)$, and, for every $\nu > 0$, the group ${}_{p^\nu}G'$ is finite over S' while ${}_{p^\nu}H'$ is separated over S' .*

Under these assumptions, u is an immersion.

Indeed, the finiteness of ${}_pG'$ over S' implies the finiteness of its special fiber over $\kappa(s)$. Hence $G'_0 \otimes_{\kappa(s)} \overline{\kappa(s)}$ contains no subgroup isomorphic to the additive group, and the hypotheses of Corollary 7.5 apply.

Remark 7.10. Examples of M. Raynaud (XVI, 1.1(a),(b)) show that in condition (c) one cannot omit either the finiteness of the ${}_{p^\nu}G'$ over S' or the separatedness of the ${}_{p^\nu}H'$ over S' .

We now seek conditions under which u is a closed immersion. Retain the hypotheses preceding Proposition 7.9, but no longer suppose that u is surjective; assume only that it is an isomorphism on generic fibers. We already know that $u : G \rightarrow H$ is an open immersion. The same is therefore true of $u_0 : G_0 \rightarrow H_0$, whose image contains the identity component $(H_0)^0$. Since G_0 is smooth over k and has no additive component after extending to $\bar{k}$, the same is true of H_0 . We use:

Lemma 7.11. *Let H be a commutative algebraic group over a field k , and suppose that $H \otimes_k \bar{k}$ contains no subgroup isomorphic to $\mathbf{G}_a$. If $n = \deg(H/H^0)$, then*

$${}_nH \longrightarrow H/H^0$$

is surjective.

We may assume that k is algebraically closed, in which case this follows from the familiar fact that $H^0(k)$ is a divisible group.

Return to $u : G \rightarrow H$, and suppose that the ${}_nG$, $n > 0$, are proper over S , while the ${}_nH$ are separated over S . The ${}_nH$ are also flat over S . Indeed, it is enough to check flatness at the points above s , and hence to check that $n \operatorname{id}_{H_0}$ is flat. As noted above, this is equivalent to $(H_0)^0$ being smooth⁴⁰ over k and having no $\mathbf{G}_a$ -component over $\bar{k}$. Since $(H_0)^0 = (G_0)^0$, this follows from the corresponding hypothesis on G_0 .

Because H is separated over S , so are the ${}_nH$. The morphism ${}_nG \rightarrow {}_nH$ is proper and therefore has closed image. Its image contains the generic fiber of ${}_nH$, because $u_{s'} : G_{s'} \rightarrow H_{s'}$ is an isomorphism. Since ${}_nH$ is flat over S , it is the schematic closure of its generic fiber. Hence ${}_nG \rightarrow {}_nH$, and therefore ${}_nG_0 \rightarrow {}_nH_0$, is surjective. Together with

⁴⁰Editor's note: the terminology has been updated, replacing "simple" by "smooth"; see, for example, A. Grothendieck's note (1) in SGA 1, Exposé II.

$G_0 \supset (H_0)^0$, Lemma 7.11 now implies that $G_0 \rightarrow H_0$ is surjective. Hence u is surjective. We obtain:

Proposition 7.12. *With the notation of 7.9, suppose that conditions (a) and (b) hold, together with the stronger condition (c') obtained from (c) by requiring, for every integer $n > 0$, not only those of the form p^ν , that ${}_n G'$ be finite over S' and ${}_n H'$ separated over S' . Then u is a closed immersion.*

Remark 7.13.

- (a) The separatedness hypothesis on the ${}_n H$ easily implies that H itself is separated over S . This is also formally contained in Proposition 7.12, by taking G to be the unit S -group. If S is locally Noetherian, in Proposition 7.9 it is enough to assume that H is locally of finite type over S , rather than of finite type; the proof applies verbatim. For Proposition 7.12, one also assumes that the fibers of H are of finite type.
- (b) Using Proposition 7.12, it is not difficult to prove that if $u : G \rightarrow H$ is a monomorphism of group preschemes of finite presentation over S , G is diagonalizable—or, more generally, of multiplicative type—and H is separated over S , then u is a closed immersion. This is a satisfactory generalization of the first conclusion in 5.7. If G is smooth over S , apply Proposition 7.12 directly; the general case reduces readily to this one. If H is not assumed separated over S , one can still show that u is an immersion; for a torus G , this also follows from the material that comes next.
- (c) If G is assumed to have connected fibers in Proposition 7.9, the separatedness of ${}_{p^\nu} H'$ over S' may be omitted from condition (c). Indeed, after the reductions in the proof, H may be assumed flat over S and u bijective. Thus H has connected fibers, and Raynaud's theorem (VI_B, 5.5) implies that H is separated over S .⁴¹

Bibliography

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⁴¹Editor's note: the attribution, which appeared in SGAD 1965, should be made explicit in Exposé VI_B, §5, together with the changes between SGAD and Lecture Notes 151.

⁴²Editor's note: additional references cited in this exposé.